

ECO 362: Financial Investments

Liquidity

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Theory of no arbitrage

- Most of finance theory built on the assumption of no arbitrage.
 - CAPM.
 - Arbitrage Pricing Theory.
 - Term structure model.
 - Derivatives pricing and Black-Scholes model.
- Benchmark model with powerful predictions.
- Hard to find arbitrage opportunities. Even sophisticated investors like mutual funds and hedge funds have difficulty beating the market.

Liquidity

- Liquidity: How easy is an asset to trade.
- Two assets with identical cash flows can trade at different prices because of liquidity.
- Liquidity spread increases when arbitrage is more constrained.
- Arbitrage can be constrained by
 - Leverage constraints.
 - Short-sales constraints.
 - Risk of noise traders who move prices in the wrong direction.

Arbitrage in the Treasury market?

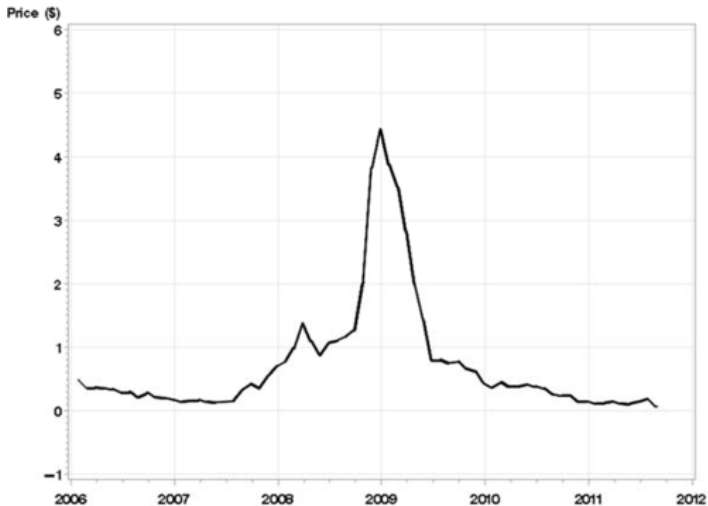
- There are small deviations from no arbitrage because of a spread between on-the-run (new) and off-the-run (old) bonds.
- For example,
 - New 10-year note with 4% coupon.
 - Old bond (30-year bond issued 20 years ago) with 11% coupon.
- Through STRIPS, the principal and coupons on the old bond can be separately traded.
- Define price difference as

$$\text{New note} - \left(\frac{4}{11} \text{Old bond coupon} + \text{Old bond principal} \right)$$

- Theory of no arbitrage predicts a price difference of 0.

- In practice, the price difference is positive (new bonds are more expensive) and small.
- In extraordinary times, the price difference can be large.
- Which is worth more?
 - Two iPhones (identical model), one is new and the other certified/refurbished.
 - Two \$100 bills, one is crisp and the other wrinkled.
- Are Treasury notes/bonds more like iPhones or \$100 bills?

Average price difference between new notes and old bonds



Source: Musto et al. (2018, *Review of Financial Studies*)

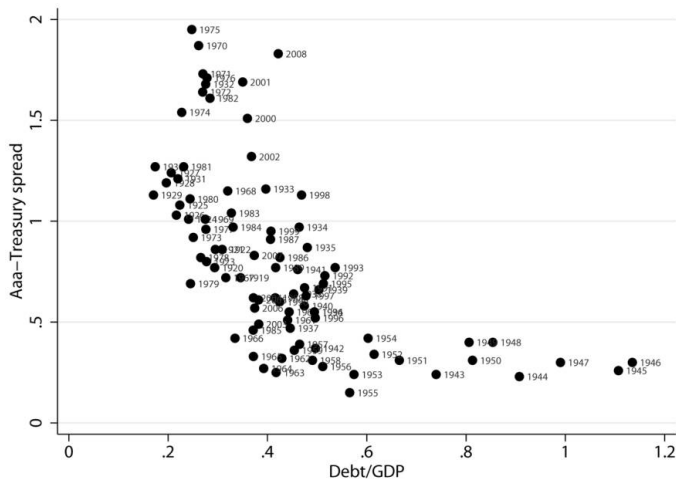
What happened during the financial crisis?

- Old bonds are less liquid (i.e., trade less frequently) because they are held by long-term investors like insurance companies who hardly trade.
- In normal times, the liquidity spread between old bonds and new notes is small.
- During the 2008 financial crisis, the average price difference between new notes and old bonds peaked at about 5%.
- Many speculators were sidelined because of large drops in their wealth.
- Many assets appeared to be mispriced during this extraordinary time, so 5% may have been small in the overall scheme of things.

Aaa corporate versus Treasuries

- Based on cash flows alone, Aaa corporate bonds should be a close substitute for Treasuries.
- However, Treasuries are more liquid.
 - Trade more frequently.
 - Easier to use as collateral.
- The liquidity spread depends on the supply of Treasuries.
 - Low Treasury supply implies higher prices or lower yields.
 - Consequently, the Aaa-Treasury yield spread is higher.

Downward sloping demand for Treasuries?



Source: Krishnamurthy and Vissing-Jorgensen (2012, *Journal of Political Economy*)

Example 1: Royal Dutch/Shell Group

- An example of “twin shares”.
- Based on a 1907 merger agreement, Royal Dutch shares receive 60% and Shell shares receive 40% of all cash flows.
- Royal Dutch trades in Amsterdam, Shell trades in London, and both have ADRs that trade in the US.
- Define

$$\text{Deviation} = \frac{\text{Royal Dutch}}{1.5 \times \text{Shell}} - 1$$

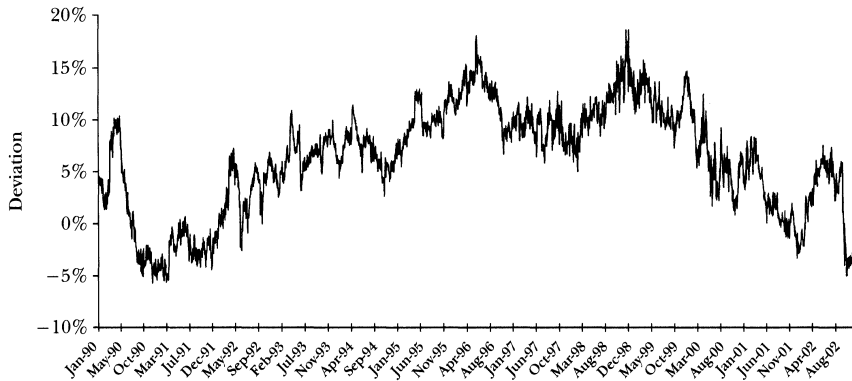
- Law of one price implies that the deviation should be 0.

Pricing of Royal Dutch relative to Shell

Figure 1

Pricing of Royal Dutch Relative to Shell

(deviation from parity)



Source: Lamont and Thaler (2003, *Journal of Economic Perspectives*)

Why did the law of one price fail?

- Why don't mutual funds buy the cheaper Shell shares instead of the more expensive Royal Dutch shares?
- Royal Dutch was in the S&P 500 index. So S&P 500 index funds can hold Royal Dutch but not Shell.
- Thus, there is more “demand” for Royal Dutch.
- On July 10, 2002, S&P announced that it was dropping all foreign stocks from the index (including Royal Dutch). The deviation fell from 6% to 1% on announcement.

Example 2: Equity carve-out of Palm

- 3Com decided to spin off its Palm division.
 1. Sell 5% of Palm shares in an IPO.
 2. In 6 months, distribute the remaining 95% to 3Com shareholders. Each 3Com shareholder will receive 1.5 shares of Palm.
- Define

$$\text{Stub} = 3\text{Com} - 1.5 \times \text{Palm}$$

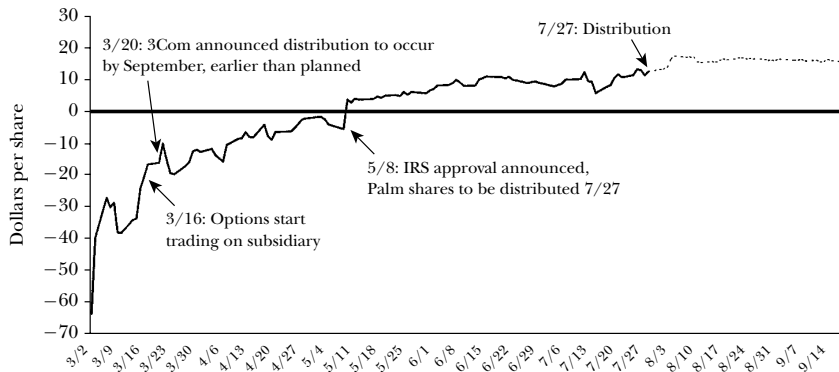
- Stub is the implied value of 3Com without the Palm division.
- The law of one price (and limited liability) predicts that the stub value cannot be negative.
- On the day of Palm IPO,

$$\text{Stub} = \$82 - 1.5 \times \$95 = -\$60.50$$

Pricing of 3Com relative to Palm

Figure 3

3Com/Palm Stub, 3/2/00–9/18/00



Source: Lamont and Thaler (2003, *Journal of Economic Perspectives*)

Why did the law of one price fail?

- Speculators would have to short Palm in order to arbitrage.
- Palm shares were difficult to short (i.e., borrow) because they were mostly held by retail investors.
- Does not explain why so many retail investors bid up Palm when they could have bought 3Com at a bargain.
- Several other examples of negative stub values during the dot-com bubble: Creative/UBID, HNC/Retek, Daisytek/PFSWeb, Metamor/Xpedior, and Methode/Stratos.

Arbitrage is risky

The market can stay irrational longer than you can stay solvent. (John Maynard Keynes)

- Even if speculators can short the over-priced stock, they are not guaranteed to make money in the short run.
- Mispricing can get worse because of noise traders and momentum traders.
- For hedge funds and other institutional investors, their clients may start withdrawing money after short-term losses.
- You may be able to make money in the long run if your clients are patient.

Margin accounts

- Textbook arbitrage does not require any money upfront.
- Implicit is an infinite leverage assumption: You can borrow as much as you would like.
- In practice, Regulation T puts restrictions on leverage in margin accounts.
- On long positions, you can only borrow up to 50% of the share price.
- On short positions, you must put an extra 50% of the share price in a margin account to cover potential losses.

Arbitrage positions in practice

- Suppose you want to arbitrage the difference in returns between Shell (R_I) and Royal Dutch (R_S).
- Buy \$1 of Shell by putting down \$0.50 and borrowing \$0.50 at an interest of R_b . The return is

$$\frac{(1 + R_I) - 0.5(1 + R_b)}{0.5} - 1 = 2R_I - R_b$$

- Borrow \$1 of Royal Dutch, sell it and put the proceeds together with an additional \$0.50 in a margin account that earns an interest of R_b . The return is

$$\frac{1.5(1 + R_b) - (1 + R_S)}{0.5} - 1 = 3R_b - 2R_S$$

- Adding the long and short positions, the portfolio return is

$$R_p = 0.5(2R_l - R_b) + 0.5(3R_b - 2R_s) = R_b + R_l - R_s$$

- Define the excess return as

$$R_p - R_b = R_l - R_s$$

- Define the Sharpe ratio as

$$\frac{\mathbb{E}[R_p - R_b]}{\sigma(R_p - R_b)} = \frac{\mathbb{E}[R_l - R_s]}{\sigma(R_l - R_s)}$$

- This strategy is expected to be profitable $\mathbb{E}[R_l - R_s] > 0$ but has significant risk $\sigma(R_l - R_s) > 0$.

Summary

- Liquidity: How easy is an asset to trade.
- In normal times, liquidity spreads are small because of arbitrage mechanisms.
- In bad times, liquidity spreads can spike.